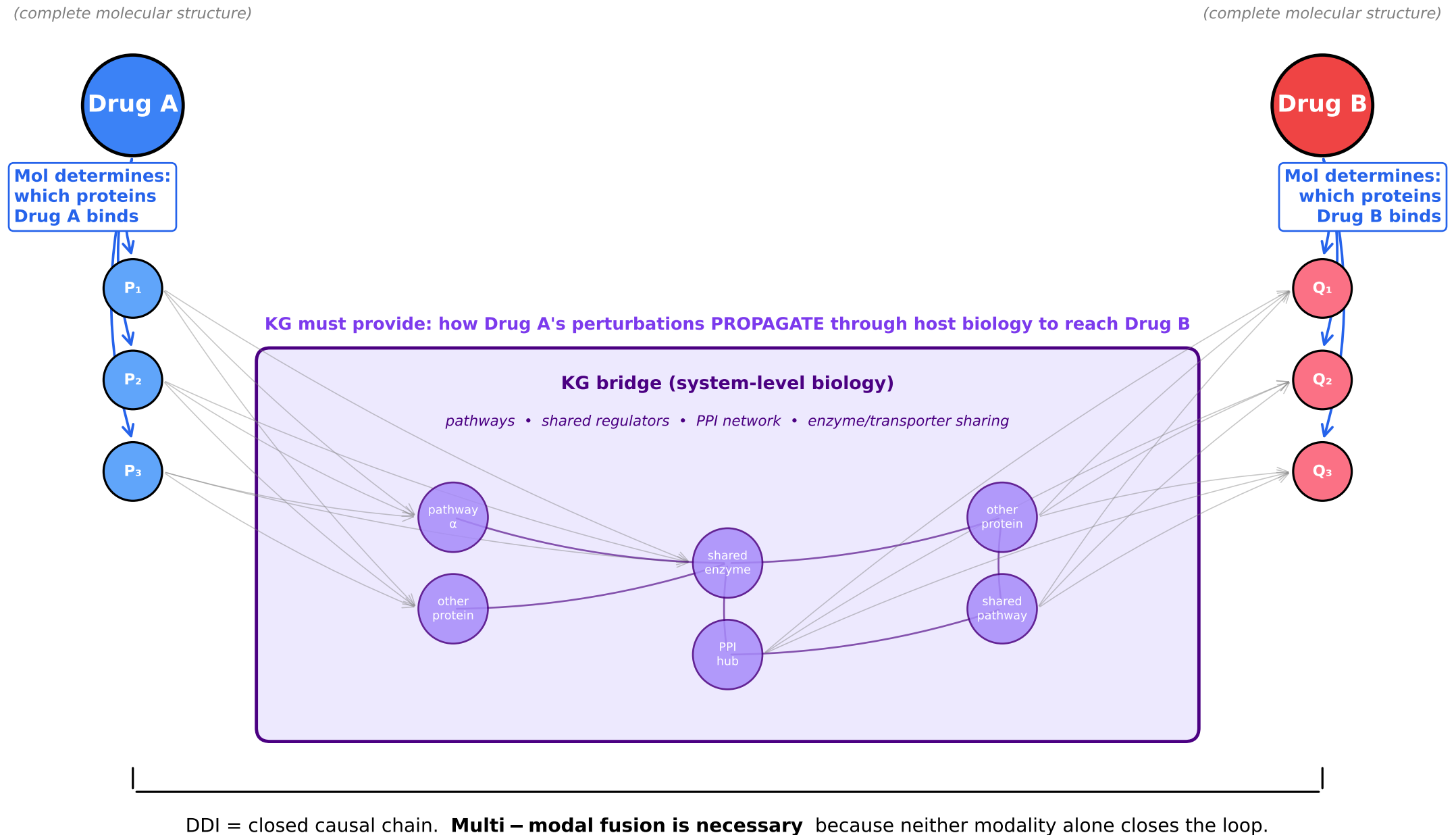


Figure 1. Multi-modal is necessary for DDI prediction.

Mol determines each drug's binding profile (endpoints). KG determines the system-level bridge that links the two drugs into a single causal chain.



Drug A's molecular structure determines which proteins it binds (left). Drug B's molecular structure determines its own binding profile (right). But whether the two drugs INTERACT depends on how their perturbations compose in host biology — a property of the protein/pathway network (KG), not of either drug alone. In cold-start, both drugs are unseen at training; the bridge cannot be inferred from Drug A's molecular structure alone, making multi-modal fusion necessary by construction.